Simple Stochastic Order-Book Model of Swarm Behavior in Continuous Double Auction

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Abstract

We present a simple stochastic order-book model for investors' swarm behaviors seen in the continuous double auction mechanism, which is employed by major global exchanges. Our study shows a characteristic called "fat tail" is seen in the data obtained from our model that incorporates the investors' swarm behaviors. The result demonstrated that one of the reasons the trend following of price occurs is that orders temporarily swarm on the order book in accordance with past price trends.

Keyword: stochastic order-book model, continuous double auction, swarm behavior

Numerous stocks and derivatives are traded on exchanges around the world. For the traders, it is crucial to estimate the profitability and downside risk of financial products they trade in. R. N. Mantegna and H. E. Stanely have pointed out, prices actually do not ideally move following a normal distribution [2]. We think one of the factors of large-scale price movement is connected with certain swarm behaviors of investors. Y. Hashimoto et al. used actual market data and indicated that the market price is dependent on past price trends [3]. We present a simple stochastic order-book model for investors' swarm behaviors seen in the continuous double auction mechanism.

Basis of our model was created by improving the no-intelligence order-book model S. Maslov introduced [1]. Moreover, we present a simple stochastic order-book model that incorporates the swarm behaviors of investors. Specifically, two swarm behaviors are incorporated in the model; one is investors' behavior to follow a trend in the historical price movement, and another is investors' behavior to send orders that contradict a trend in the historical price movement.

We analyzed the extent to which the price continuously moved in a single direction. We obtained price data from our simulation and plotted a frequency distribution of the absolute value of the draw down and draw up in the price. The absolute values of the draw down and draw up are called the draw size. In particular, a fat tail could be observed from the distribution of the draw size obtained from the model incorporating the swarm behavior of investors that follow the trend of the past. We saw the power law in the tail of this cumulative frequency distribution.

This shows that, compared to a market that assumes price movement as a random walk, larger continuous price movements occur more frequently. Numerous investor's behaviors interact and structure themselves as a swarm behavior and create a phenomenon similar to significant price decline. In fact, the result demonstrated that one of the reasons the trend following of price occurs is that the orders temporarily swarm on the order book in accordance with past price trends.

References

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